WORKED SOLUTION

# STEP I 2018, Question 12

Probability and Statistics • Sampling With Replacement

**STEP I 2018, Question 12**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s1-q12

## The key idea

Write each count of heads as a sum of two indicator variables, one for each toss. The expectation of a sum is the sum of expectations whether or not the tosses are independent, so the means come out easily. The variances turn on a single number, the probability that *both* tosses show heads, and that is exactly where drawing with replacement and drawing without replacement differ.

## Solution

Throughout, let $p = \frac{1}{3}(p_1 + p_2 + p_3)$ and

$$q = p_1p_2 + p_2p_3 + p_3p_1.$$

We assume that each coin is equally likely to be chosen and that, once the coins have been chosen, their tosses are independent.

### Part (i)

Let $C_k$ be the event that coin $k$ is chosen, so $\mathrm{P}(C_k) = \frac{1}{3}$. By the law of total probability,

$$\mathrm{P}(\text{head}) = \sum_{k=1}^{3} \mathrm{P}(C_k)\,\mathrm{P}(\text{head} \mid C_k) = \tfrac{1}{3}(p_1 + p_2 + p_3) = p.$$

### Part (ii)

Let $X_1$ and $X_2$ be the indicators of a head on the first and second toss, so $N_1 = X_1 + X_2$. Because the first coin is returned, the second toss happens in exactly the same situation as the first, and by part (i) $\mathrm{P}(X_1 = 1) = \mathrm{P}(X_2 = 1) = p$. Hence

$$\mathrm{E}(N_1) = \mathrm{E}(X_1) + \mathrm{E}(X_2) = 2p.$$

For the variance we need $\mathrm{P}(X_1 = 1,\, X_2 = 1)$. The two choices are independent, each of the 9 ordered pairs of coins $(j,k)$ has probability $\frac{1}{9}$, and given the pair the chance of two heads is $p_j p_k$. So

$$\mathrm{P}(X_1 = 1,\, X_2 = 1) = \frac{1}{9}\sum_{j=1}^{3}\sum_{k=1}^{3} p_j p_k = \left(\frac{p_1+p_2+p_3}{3}\right)^2 = p^2 = \mathrm{P}(X_1 = 1)\,\mathrm{P}(X_2 = 1).$$

The same factorisation works for the other outcomes, so $X_1$ and $X_2$ are independent and $N_1 \sim \mathrm{B}(2,p)$. Directly: since $X_i^2 = X_i$,

$$\begin{aligned}\mathrm{E}(N_1^2) &= \mathrm{E}(X_1) + \mathrm{E}(X_2) + 2\,\mathrm{E}(X_1X_2) = 2p + 2p^2,\\ \mathrm{Var}(N_1) &= 2p + 2p^2 - (2p)^2 = 2p(1-p),\end{aligned}$$

as required.

**Part (iii)**

Now pick the two coins one after the other without replacement, and let $Y_1$, $Y_2$ be the indicators of a head on the first and second coin picked, so $N_2 = Y_1 + Y_2$. Each coin is equally likely to be the first one picked, and by symmetry each is equally likely to be the second, so as in part (i) $\mathrm{P}(Y_1 = 1) = \mathrm{P}(Y_2 = 1) = p$ and

$$\mathrm{E}(N_2) = 2p.$$

? **Ask yourself** Can we reuse the calculation of P(both heads) from part (ii)?

No: the same coin can no longer be picked twice. There are now 6 equally likely ordered pairs $(j,k)$ with $j \neq k$, and each unordered pair $\{j,k\}$ appears twice, so

$$\mathrm{P}(Y_1 = 1,\, Y_2 = 1) = \frac{1}{6}\sum_{j\neq k} p_j p_k = \frac{2q}{6} = \frac{q}{3}.$$

In general this is not $p^2$, so $Y_1$ and $Y_2$ are not independent. Using $Y_i^2 = Y_i$ again,

$$\begin{aligned}\mathrm{E}(N_2^2) &= \mathrm{E}(Y_1) + \mathrm{E}(Y_2) + 2\,\mathrm{E}(Y_1Y_2) = 2p + \tfrac{2}{3}q,\\ \mathrm{Var}(N_2) &= 2p + \tfrac{2}{3}(p_1p_2 + p_2p_3 + p_3p_1) - 4p^2.\end{aligned}$$

**Part (iv)**

Subtracting,

$$\mathrm{Var}(N_1) - \mathrm{Var}(N_2) = (2p - 2p^2) - (2p + \tfrac{2}{3}q - 4p^2) = 2p^2 - \tfrac{2}{3}q.$$

Now $9p^2 = (p_1+p_2+p_3)^2 = p_1^2 + p_2^2 + p_3^2 + 2q$, so

$$\begin{aligned}2p^2 - \tfrac{2}{3}q &= \tfrac{2}{9}(p_1^2 + p_2^2 + p_3^2 + 2q) - \tfrac{6}{9}q = \tfrac{2}{9}(p_1^2 + p_2^2 + p_3^2 - p_1p_2 - p_2p_3 - p_3p_1)\\ &= \tfrac{1}{9}\Big[(p_1-p_2)^2 + (p_2-p_3)^2 + (p_3-p_1)^2\Big],\end{aligned}$$

the last step because expanding the three squares gives $2(p_1^2 + p_2^2 + p_3^2) - 2q$.

A sum of squares of real numbers is non-negative, so $\mathrm{Var}(N_2) \leq \mathrm{Var}(N_1)$. Equality holds if and only if all three squares vanish, that is, if and only if $p_1 = p_2 = p_3$. This completes the proof.

As a by-product we get a tidy form for the answer to part (iii):

$$\mathrm{Var}(N_2) = 2p(1-p) - \tfrac{1}{9}\Big[(p_1 - p_2)^2 + (p_2 - p_3)^2 + (p_3 - p_1)^2\Big].$$

## Checks

- If $p_1 = p_2 = p_3 = p$, every coin is the same, so it cannot matter how the coins are chosen: both counts are $\mathrm{B}(2, p)$ and the variances agree, matching the equality case.
- Take $p_1 = \frac{1}{2}$, $p_2 = \frac{1}{3}$, $p_3 = \frac{1}{6}$, so $p = \frac{1}{3}$ and $q = \frac{11}{36}$. Then $\mathrm{Var}(N_1) = \frac{4}{9}$ and $\mathrm{Var}(N_2) = \frac{2}{3} + \frac{11}{54} - \frac{4}{9} = \frac{23}{54}$, which is $\frac{1}{54}$ less, agreeing with $\frac{1}{9}\left(\frac{1}{36} + \frac{1}{36} + \frac{1}{9}\right) = \frac{1}{54}$. Listing all the outcomes by hand gives the same values.
- Extreme case $p_1 = p_2 = 1$, $p_3 = 0$: in part (iii) exactly one of the two chosen coins is a sure head except when coins 1 and 2 are both chosen, so $N_2$ is 2 with probability $\frac{1}{3}$ and 1 otherwise, giving variance $\frac{2}{9}$. The formula gives $2p + \frac{2}{3}q - 4p^2 = \frac{4}{3} + \frac{2}{3} - \frac{16}{9} = \frac{2}{9}$.

## Common slips

- Treating the two tosses in part (iii) as independent and getting $2p(1-p)$ again; the whole point of part (iv) is that they are not.
- Writing $\mathrm{Var}(N_1)$ as an average of the variances $2p_k(1-p_k)$ for each coin; that ignores the extra variability from the random choice of coin, and in part (ii) a different coin may be used for each toss.

## Beyond STEP

The difference between parts (ii) and (iii) is that of sampling with and without replacement. Sampling without replacement makes the draws negatively correlated, which reduces the variance; the same effect gives the "finite population correction" in survey sampling and is why the hypergeometric distribution has a smaller variance than the binomial with the same mean. The algebra in part (iv) is the identity $\sum p_k^2 - \sum_{j<k} p_j p_k = \frac{1}{2}\sum_{j<k}(p_j - p_k)^2$, a close cousin of the fact that a variance is never negative. Blitzstein and Hwang, *Introduction to Probability*, treats indicator variables, covariance and the hypergeometric distribution in exactly this spirit.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.